



Print And Build Your Own Full-Scale HUMANOID ROBOTIC ARM

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Humanoid robotics is gaining popularity, though a common misconception remains that it is complex and expensive. This system is designed to simplify entry into the field, featuring a life-size humanoid robotic arm with eight degrees of freedom: two at the shoulder, one at the elbow, and one for each finger.

The design emphasises simplicity in both mechanical and electronic aspects. Forming part of a larger humanoid robot named VulcanV1, the arm can be controlled using

an Arduino with dedicated code to manage each joint and degree of freedom. Servo motors at every joint enable precise angular movement across all axes.

The system has applications in full-scale humanoid robots, cobots, and intelligent robotic arm designs. The entire setup is custom-designed by the author and can be readily 3D printed, assembled, and operated using the required components and software.

The Author's prototype in Fig. 1 shows a life-size humanoid robotic arm with articulated joints and wiring layout.

This arm consists of shoulder, elbow, wrist, and finger sections, each driven by servo motors that provide precise angular control. Electrical wiring runs internally through the arm segments, connecting all actuators

to the central controller. When powered, the Arduino sends control signals to individual servos based on programmed instructions. Each joint moves independently, allowing coordinated motion similar to a human arm. The glowing highlights indicate active joint regions where rotation or movement occurs. The fingers are capable of gripping and releasing objects through synchronised servo action. Overall, the system converts programmed commands into mechanical motion for accurate positioning and manipulation. This makes it suitable for humanoid robots, assistive devices, and collaborative robotic applications.

This system requires 3D-printed structural parts along with electronic components for servo control. The system can be driven using either an Arduino Mega or an Arduino Uno paired with a dedicated servo driver module. Depending on the load and motion requirements of the arm, three types of servo motors are selected to ensure adequate torque and smooth operation.

For testing and operation, a 12V

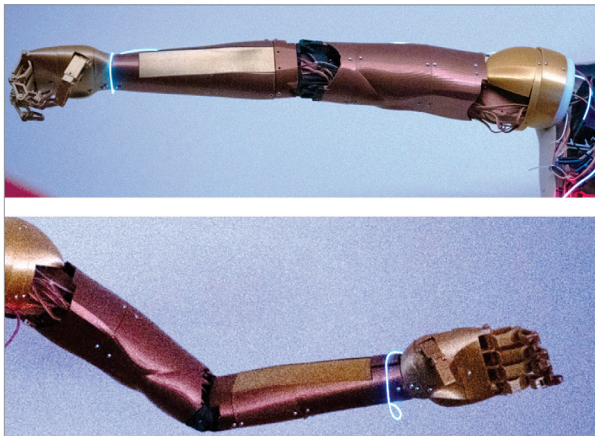


Fig. 1: Author's prototype

TABLE 1
ELECTRONIC COMPONENTS

Item	Description	Quantity	Specifications/Notes
Arduino board	Microcontroller board	1	≥ 8 PWM pins (Arduino Mega recommended if no driver is used)
PCA9685 servo driver (optional)	PWM expansion module	1	16-channel
High-torque servo (25kg-cm)	Servo motor	1	4.8V-6.8V, ~40mm×20mm×40.5mm
High-torque servo (DH03-X/ASME-MRB)	Heavy-duty servo motor	2	180kg-cm–360kg-cm, ~95.5mm×60.5mm×102.6mm
Micro servo (SG90/MG90S)	Mini servo motor	5	~22.8mm×12.2mm×28.5mm
5V power supply	Regulated supply	1	20A
12V/24V power supply	Motor supply	1	30A (24V recommended)
Servo extension cables	Male-female cables	15	30cm
14AWG wires (red and black)	Power wiring	2m each	High-current connections
3-prong power cable	AC input cable	1	Spare